

# Customer Churn Analysis

## Problem Statement (12) | SQL + Python Pandas + Power BI

### 1. Problem Statement

A subscription-based business wants to understand why customers churn and identify which customer segments are most at risk. The analysis will track churn rate by plan and tenure, compare revenue from churned and retained customers, and explore how usage patterns, complaints, and tenure relate to churn. The final solution will present churn trends, cohort analysis, and a list of at-risk customers through an interactive Power BI dashboard.

### 2. Project Objectives

- Measure overall customer churn and retention.
- Compare churn across subscription plans and tenure groups.
- Compare revenue from churned and retained customers.
- Study the relationship between usage, complaints, tenure and churn.
- Identify customer groups with higher churn risk.
- Perform structured analysis using SQL.
- Perform deeper EDA and customer analysis using Python Pandas.
- Build an interactive Power BI dashboard and communicate key findings.

### 3. Detailed Task-wise Plan

| Phase   | Tool          | Tasks                                                                                                                          |
|---------|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| Task 1  | SQL           | Create the database/table structure and load the dataset; inspect records and validate row counts, NULL values and duplicates. |
| Task 2  | SQL           | Perform basic customer analysis using SELECT, WHERE, DISTINCT, ORDER BY, GROUP BY and aggregate functions.                     |
| Task 3  | SQL           | Calculate total customers, churned customers, retained customers and overall churn rate.                                       |
| Task 4  | SQL           | Analyze churn rate by subscription plan and compare customer counts across plans.                                              |
| Task 5  | SQL           | Analyze churn rate by customer tenure groups and identify higher-churn tenure segments.                                        |
| Task 6  | SQL           | Compare revenue from churned and retained customers and calculate relevant revenue summaries.                                  |
| Task 7  | SQL           | Analyze average usage, complaints and other available behavioral fields for churned versus retained customers.                 |
| Task 8  | SQL           | Use JOINS if multiple tables are provided to combine customer, subscription, usage, complaint and revenue information.         |
| Task 9  | SQL           | Use subqueries to build churn calculations and identify customers or segments above selected churn thresholds.                 |
| Task 10 | SQL           | Use CASE expressions to create customer, tenure and risk categories where appropriate.                                         |
| Task 11 | SQL           | Use window functions such as RANK, DENSE_RANK, ROW_NUMBER or running calculations where supported by the dataset.              |
| Task 12 | SQL           | Create monthly or period-based churn trends when date fields are available and compare changes over time.                      |
| Task 13 | SQL           | Prepare a final analytical query/view containing the fields required for Python and Power BI.                                  |
| Task 14 | Python Pandas | Import the SQL dataset and inspect shape, columns, data types, missing values, duplicates and unique categories.               |
| Task 15 | Python Pandas | Clean and transform data using Pandas: handle missing values, duplicates, incorrect data types and inconsistent categories.    |

| Phase   | Tool          | Tasks                                                                                                                                                       |
|---------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task 16 | Python Pandas | Create derived columns such as tenure groups, usage groups and other analysis-ready categories based on available fields.                                   |
| Task 17 | Python Pandas | Perform descriptive statistics for numerical variables such as tenure, usage, complaints and revenue.                                                       |
| Task 18 | Python Pandas | Analyze churn distribution and calculate churn percentages using groupby, aggregation and pivot-style analysis.                                             |
| Task 19 | Python Pandas | Compare churned versus retained customers across plan, tenure, usage, complaints and revenue.                                                               |
| Task 20 | Python Pandas | Perform correlation analysis between numerical variables and churn-related indicators where appropriate.                                                    |
| Task 21 | Python Pandas | Analyze churn trends over time if date fields are available; identify periods with higher or lower churn.                                                   |
| Task 22 | Python Pandas | Perform customer segmentation using available behavioral, tenure and usage information and profile each segment.                                            |
| Task 23 | Python        | Create Matplotlib/Seaborn visualizations for churn by plan, tenure, usage, complaints and revenue.                                                          |
| Task 24 | Python        | Create churned-versus-retained comparison charts and identify the strongest observable patterns.                                                            |
| Task 25 | Python Pandas | Create an at-risk customer analysis using available churn indicators and produce a customer-level output for Power BI.                                      |
| Task 26 | Python Pandas | Validate important Python results against SQL results and prepare the final analysis dataset.                                                               |
| Task 27 | Power BI      | Import the prepared dataset, create the data model, and build KPI cards for total customers, churned customers, retained customers, churn rate and revenue. |
| Task 28 | Power BI      | Create visuals for churn by plan, tenure, usage and complaints, with slicers and interactive filtering.                                                     |
| Task 29 | Power BI      | Create churn trend and cohort views when the available date fields support them.                                                                            |
| Task 30 | Power BI      | Create an at-risk customer table/list and finalize dashboard formatting and interactions.                                                                   |
| Task 31 | Final         | Summarize key findings, business insights and recommendations; prepare the project report and presentation.                                                 |

## 4. Final Deliverables

- SQL file containing basic and advanced analytical queries
- Python Pandas notebook covering cleaning, EDA, segmentation and visualizations
- Power BI interactive dashboard
- Project report